

Female reproductive tract module validation report

Levonorgestrel simulations

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i Note

This report has been created by running the simulations. The results have been stored in folder ' C:/Projects/Female-Reproductive-Tract-PBPK-Module/Reports/Results/SimulationResults/2025-10-06 17-09 '.

1 Introduction

This report presents the validation of the female reproductive tract module using Levonorgestrel simulations. The module is designed to simulate the pharmacokinetics of drug targetting tissues in the female reproductive tract, and distribution of the drug upon local administration (e.g., vaginal, intrauterine or cervical).

Levonorgestrel is a synthetic progestogen widely used in various hormonal contraceptive formulations. A PBPK model for oral administration of levonorgestrel was obtained from literature ([Cicali et al. 2021](#)), and extended for intrauterine (IUD) administration.

2 Compound: Levonorgestrel 1

2.1 Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	14 mg/l	Other-Assumption-Biorelevant solubility estimated using Simcyp	Estimated FaSSIF	True
Reference pH	6.5	Other-Assumption-Biorelevant solubility estimated using Simcyp	Estimated FaSSIF	True
Lipophilicity	4 Log Units	Database-Manual Fit	logP	True
Fraction unbound (plasma, reference value)	1	Other-Other-Binding kinetics operating in mechanistic mode	fu	True
Permeability	0.31 cm/min		Parameter Estimation	False
Specific intestinal permeability (transcellular)	0.0002 cm/min		Fitted	True
Is small molecule	Yes			
Molecular weight	312.44 g/mol			
Plasma protein binding partner	Unknown			
Density (drug)	1.2 g/cm ³	Unknown		

2.2 Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

2.3 Processes

2.3.1 Systemic Process: Total Hepatic Clearance-Parameter identification

Species: Human

2.3.1.1 Parameters

Name	Value	Value Origin
Body weight	60 kg	Unknown
Hematocrit	0.41	Unknown
Fraction unbound (experiment)	0.014	
Lipophilicity (experiment)	3.65	
	Log Units	
Plasma clearance	0.1 l/h/kg	Parameter Identification

2.3.2 Metabolizing Enzyme: CYP3A4-Parameter Identification

Molecule: CYP3A4

Metabolite: Metabolite

2.3.2.1 Parameters

Name	Value	Value Origin
In vitro CL/recombinant enzyme	1.45 μ l/min/pmol rec. enzyme	Parameter Identification

2.3.3 Specific Binding: SHBG-Qi-Gui & Humpel 1990

Molecule: SHBG

2.3.3.1 Parameters

Name	Value	Value Origin
koff	17.4 1/min	
Kd	0.0005 μ M	

2.3.4 Specific Binding: ALB-Bayer report

Molecule: ALB

2.3.4.1 Parameters

Name	Value	Value Origin
koff	6.5 1/min	
Kd	6 μ mol/l	

3 Clinical Data

Data from the following studies has been used for model validation ([Cicali et al. 2021](#); [Apter et al. 2014](#); [Nilsson et al. 1982](#)):

Reference	Administration route	Dose	Clinical data	Population	Weight (kg)	Height (cm)	Age (years)	Number of par- ticipants
Cicali 2021	PO	0.09 mg	Plasma	European ICRP (female)	-	-	-	<20
Nilsson 1982	PO	0.25 mg	Plasma; Fat; Endometrium; Myometrium	European ICRP (female)	-	-	-	4
Apter 2014	Intrauterine (IUD)	20 μ g/24h	Plasma	European ICRP (female)	-	-	-	<15

4 Results

Comparison of individual simulations with observed data are presented in the following figures.

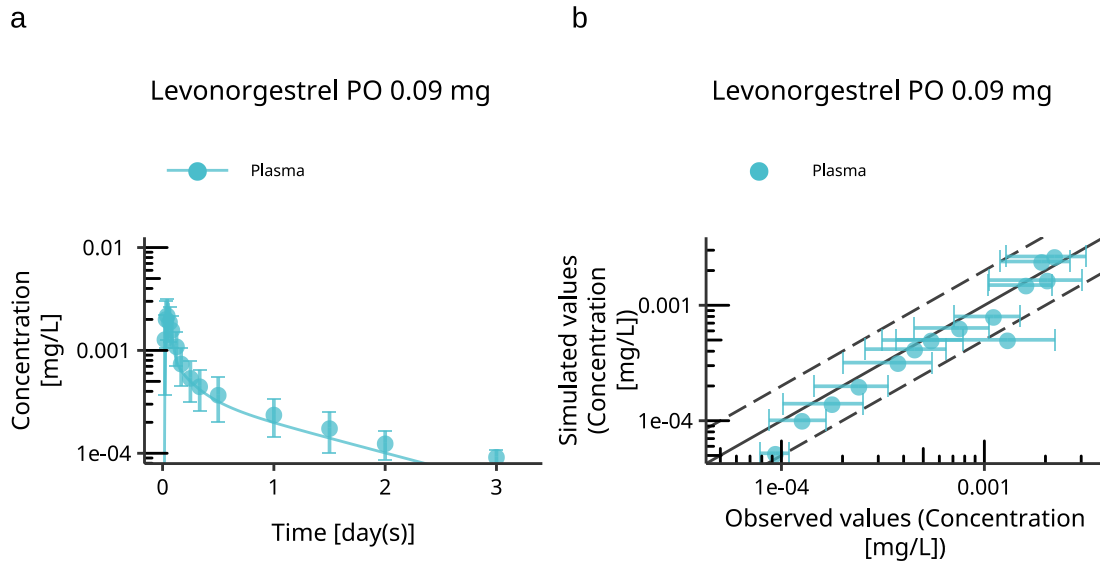
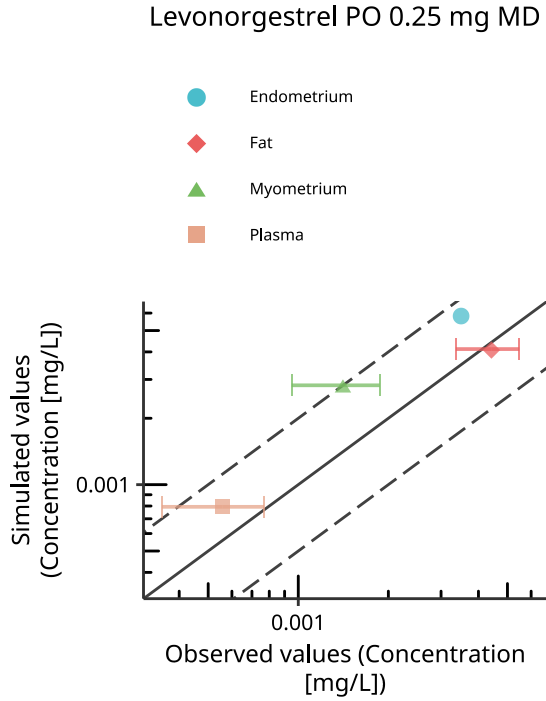


Figure 1: Concentration-time profiles (a) and goodness-of-fit plots (b) of levonorgestrel administered as a single 0.09 mg oral dose (a, b) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Cicali et al. 2021). Solid line represent unity, and dashed lines represent 2-fold deviation.

a



b

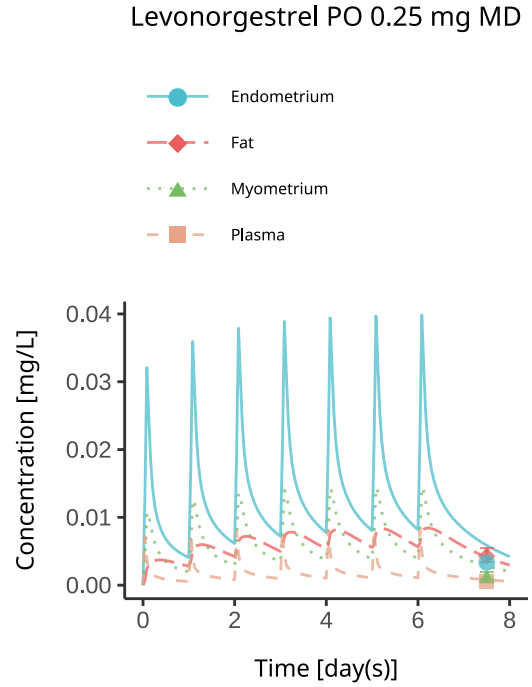


Figure 2: Concentration-time profiles (a) and goodness-of-fit plots (b) of levonorgestrel administered as a multiple 0.25 mg oral dose (a, b) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Nilsson et al. 1982). Solid line represent unity, and dashed lines represent 2-fold deviation.

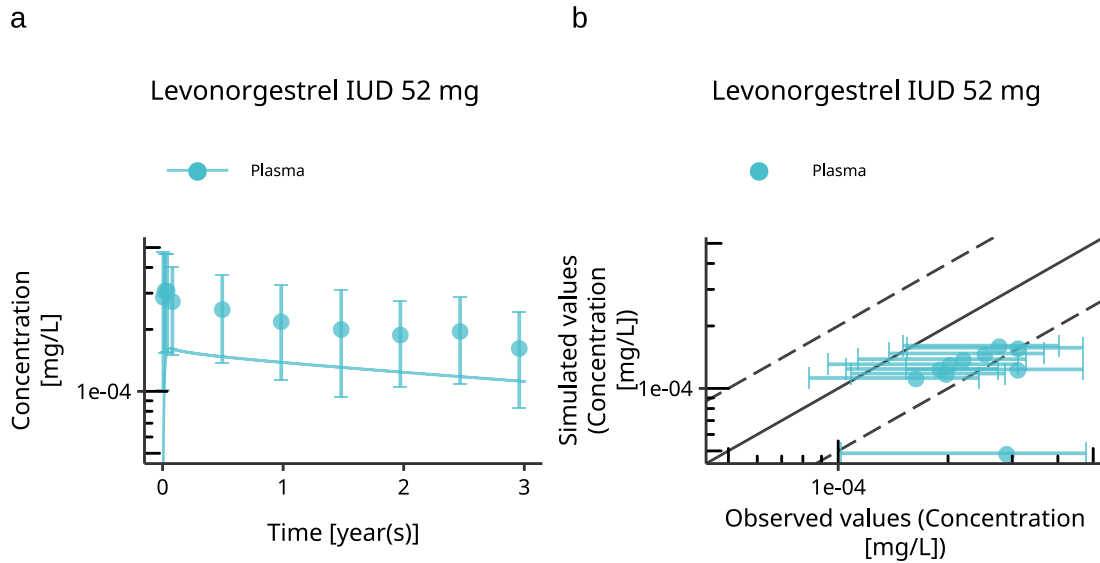


Figure 3: Concentration-time profiles (a) and goodness-of-fit plots (b) of levonorgestrel administered as a 52 mg intrauterine (a, b) administration. Lines represent predicted concentration for the mean individual. Points (errorbars) represent mean (standard deviation) observed data (Apter et al. 2014). Solid line represent unity, and dashed lines represent 2-fold deviation.

5 References

References

- Apter, Dan, Kristina Gemzell-Danielsson, Brian Hauck, Kimberly Rosen, and Christian Zurth. 2014. "Pharmacokinetics of Two Low-Dose Levonorgestrel-Releasing Intrauterine Systems and Effects on Ovulation Rate and Cervical Function: Pooled Analyses of Phase II and III Studies." *Fertility and Sterility* 101 (6): 1656–1662.e4. <https://doi.org/10.1016/j.fertnstert.2014.03.004>.
- Cicali, Brian, Karthik Lingineni, Rodrigo Cristofolletti, Thomas Wendl, Joachim Hoechel, Herbert Wiesinger, Ayyappa Chaturvedula, Valvanera Vozmediano, and Stephan Schmidt. 2021. "Quantitative Assessment of Levonorgestrel Binding Partner Interplay and Drug-Drug Interactions Using Physiologically Based Pharmacokinetic Modeling." *CPT: Pharmacometrics & Systems Pharmacology* 10 (1): 48–58. <https://doi.org/10.1002/psp4.12572>.
- Nilsson, C. G., M. Haukamaa, H. Vierola, T. Luukkainen, and P. Arcangeli. 1982. "TIS-SUE CONCENTRATIONS OF LEVONORGESTREL IN WOMEN USING A LEV-

ONORGESTREL-RELEASING IUD.” *Clinical Endocrinology* 17 (6): 529–36. <https://doi.org/10.1111/j.1365-2265.1982.tb01625.x>.